

CHOOSE THE CORRECT ANSWER

(30 POINTS)

الاجابة تكون في الجدول الموجود باسفل صفحة الغلاف

1. CIL02

(Variation of parameters method)

For the differential equation

$$x^2 y'' - 4xy' + 6y = x^2 \ln x, \quad x > 0$$

if the particular solution ;

$$y_p = (x^2)u_1 + (x^3)u_2 \quad \text{then } u_1' =$$

- A. $-\frac{\ln x}{x^2}$ B. $\frac{\ln x}{x}$ **C. $-\frac{\ln x}{x}$** D. $\frac{\ln x}{x^2}$

2. CIL01

The integrating factor that makes the differential equation;

$$(xy)dx + (5+y)dy = 0, \quad x > 0, y > 0$$

to be exact, is:

- A. $\mu(y) = y^{-1}$** B. $\mu(x) = x^{-2}$ C. $\mu(x) = x^{-1}$ D. $\mu(y) = y^{-2}$

3. CIL01

The general solution of the following differential equation;

$$y' + \frac{y}{x} = 2 \sin x^2, \quad x > 0$$

is:

- A. $y = \frac{1}{x}(-\cos x^2 + c)$** C. $y = \frac{1}{x}(\cos x^2 + c)$
B. $y = \frac{1}{x} \cos x^2 + c$ D. $y = -\frac{1}{x} \cos x^2 + c$

4. CIL01

The assumption that converts the differential equation ;

$$y' + 2xy = \frac{x}{y}, \quad x > 0, y > 0$$

to be linear is:

- A. $u = y^3$ B. $u = y^{-1}$ **C. $u = y^2$** D. $u = y^{-2}$

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5. CIL05

The general solution of the following differential equation;

$$\frac{dy}{dx} = \sqrt{xy} \quad \text{is:}$$

A. $2\sqrt{y} = \sqrt{x^3} + c$

B. $\sqrt{y} = \sqrt{x^3} + c$

☒ C. $3\sqrt{y} = \sqrt{x^3} + c$

D. $\sqrt{y} = 3\sqrt{x^3} + c$

6. CIL08

If the linear differential equation;

$$y'' + y' - 4 = f(x) \quad \text{is homogeneous then } f(x) =$$

A. 2

B. 0

C. 4

☒ D. -4

7. CIL05

The general solution of the following differential equation;

$$\frac{dy}{dx} = -1 + \sqrt{4 - (x+y)^2} \quad \text{is:}$$

A. $y = -x - 2\sin(x+c)$

B. $y = -x - 4\sin(x+c)$

C. $y = -x + 4\sin(x+c)$

☒ D. $y = -x + 2\sin(x+c)$

8. CIL01

Which of the following differential equation, is linear ?

A. $(y')^2 + 2xy = x$

☒ C. $y' + 2xy = x^2$

B. $yy' + 2xy = x^2$

D. $y' + 2xy = \sqrt{y}$

9. CIL02

(Method of undetermined coefficients)

Write the suitable assumption for the particular solution, in order to solve the following differential equation;

$$y''' - y'' = x + e^x$$

☒ A. $y_p = Ax^3 + Bx^2 + Cxe^x$

C. $y_p = Ax^3 + Bx^2 + Ce^x$

B. $y_p = Ax + B + Ce^x$

D. $y_p = Ax^2 + Bx + Ce^x$

10. CIL06

Write the non-homogeneous linear differential equation, whose general solution is

$$y = c_1 e^{10x} + c_2 e^{-10x} + \sin 2x \quad ?$$

A. $y'' + 100y = -104 \sin 2x$

C. $y'' - 100y = -96 \cos 2x$

☒ B. $y'' - 100y = -104 \sin 2x$

D. $y'' + 100y = -96 \cos 2x$

11. CILO1

The degree and the order of the differential equation;

$$\left(y^{(4)}\right)^3 + (y'')^7 = yy''' \text{ are:}$$

- A. 4 and 3 respectively
B. 7 and 4 respectively

- ☒ C. 3 and 4 respectively
D. 4 and 7 respectively

12. CILO8If the characteristic equation of the homogeneous Cauchy-Euler differential equation, is $m^3 - 1 = 0$ then the differential equation is:

A. $x^3 y''' + y = 0$

B. $x^3 y''' - y = 0$

☒ C. $x^3 y''' + 3x^2 y'' + xy' - y = 0$

D. $x^3 y''' + 3x^2 y'' + xy' + y = 0$

13. CILO2

Solve the (IVP);

$$x^2 y'' - 2xy' + 2y = 0, \quad y(1) = 2, \quad y'(1) = 4$$

A. $y = 2x^3 - x^2 + 1$

B. $y = x^4 + 1$

☒ C. $y = 2x^2$

D. $y = x^2 + 4x - 3$

14. CILO2

The general solution of the differential equation;

$$y''' - 4y' = 0$$

is:

A. $y = c_1 + c_2 e^{-2x} + c_3 x e^{-2x}$

B. $y = c_1 + c_2 e^{2x} + c_3 x e^{2x}$

☒ C. $y = c_1 + c_2 e^{2x} + c_3 e^{-2x}$

D. $y = c_1 e^{2x} + c_2 e^{-2x} + c_3 x e^{-2x}$

15. CILO6Write the homogeneous linear differential equation with constant coefficients, if the roots of its characteristic equation are $3 \pm \sqrt{2}i$?

A. $y'' - 6y' + 11 = 0$

B. $y'' - 6y' - 11 = 0$

☒ C. $y'' - 6y' + 11y = 0$

D. $y'' + 6y' + 11y = 0$

Given that :

$$\frac{dy}{dx} = 2x + 2xy^2 \quad , \quad y(0) = 0$$

then

$$y\left(\sqrt{\frac{3}{4}}\pi\right) =$$

Select one:

- ☐ a. 1
- ☐ b. $\sqrt{3}$
- ☒ c. -1
- ☐ d. $-\sqrt{3}$

[Clear my choice](#)

The assumption, $u = \frac{y}{x}$ converts the differential equation ;

$$\frac{dy}{dx} = \frac{x+y}{x-y} \text{ into}$$

Select one:

☐ a. $x \frac{du}{dx} = \frac{1-u^2}{2-u}$

☐ b. $x \frac{du}{dx} = \frac{1+u^2}{2-u}$

☐ c. $x \frac{du}{dx} = \frac{2+u^2}{1-u}$

☒ d. $x \frac{du}{dx} = \frac{1+u^2}{1-u}$

[Clear my choice](#)

If the order and the degree of the differential equation;

$$(y''')^4 + (y'')^5 + (y')^6 = e^x$$

are a and b respectively, then the order pair $(a, b) =$

Select one:

- ☒ a. $(3, 4)$
- ☐ b. $(6, 3)$
- ☐ c. $(3, 6)$
- ☐ d. $(4, 3)$

[Clear my choice](#)

The integrating factor that makes the differential equation;

$$2x dx + \left(\frac{x^2}{\sqrt{1-y^2}} + y \right) dy = 0$$

to be exact, is:

Select one:

- ☒ a. $e^{\sin^{-1} y}$
- ☐ b. $\tan^{-1} y$
- ☐ c. $\sin^{-1} y$
- ☐ d. $e^{\tan^{-1} y}$

[Clear my choice](#)

The general solution of the differential equation;

$$x \frac{dy}{dx} + 2y = 4x\sqrt{y} \quad , x > 0$$

is:

Select one:

- ☐ a. $(x + \frac{c}{x})$, where c is an arbitrary constant.
- ☐ b. $(2x + \frac{c}{x})^2$, where c is an arbitrary constant.
- ☒ c. $(x + \frac{c}{x})^2$, where c is an arbitrary constant.
- ☐ d. $(2x + \frac{c}{x})$, where c is an arbitrary constant.

[Clear my choice](#)

If $y = \sum_{n=0}^{\infty} a_n x^n$ is the series solution of the differential equation;

$$y'' - 4y = 0$$

then that produces the following recurrence relation:

Select one:

- ☐ a. $a_{n+2} = -\frac{4a_n}{(n+2)(n+1)}, n = 0, 1, 2, 3...$
- ☒ b. $a_{n+2} = \frac{4a_n}{(n+2)(n+1)}, n = 0, 1, 2, 3...$
- ☐ c. $a_{n+2} = \frac{2a_n}{(n+2)(n+1)}, n = 0, 1, 2, 3...$
- ☐ d. $a_{n+2} = -\frac{2a_n}{(n+2)(n+1)}, n = 0, 1, 2, 3...$

[Clear my choice](#)

If $y = \sum_{n=0}^{\infty} a_n x^n$ is the series solution of the following differential equation;

$$y'' + xy = 0$$

then the coefficient of x^4 , is equal to:

Select one:

- ☐ a. $-\frac{a_1}{12}$
- ☐ b. a_1
- ☐ c. $\frac{a_1}{12}$
- ☐ d. 0

The Maclaurin series corresponding to the function;

$$f(x) = x^3 e^{x^2}$$

is:

Select one:

☐ a. $\sum_{n=0}^{\infty} \frac{x^{3n+2}}{n!}$

☐ b. $\sum_{n=0}^{\infty} \frac{x^{n^3+2}}{n!}$

☐ c. $\sum_{n=0}^{\infty} \frac{x^{2n+3}}{n!}$

☐ d. $\sum_{n=0}^{\infty} \frac{x^{n^2+3}}{n!}$

If $y = \sum_{n=0}^{\infty} a_n x^n$ is the series solution of the following differential equation;

$$y'' - x^2 y = 0$$

then that implies:

Select one:

- ☐ a. $0 = a_2 = a_6 = a_{10} = a_{14} = \dots$
- ☐ b. $0 = a_1 = a_3 = a_5 = a_7 = \dots$
- ☐ c. $0 = a_0 = a_2 = a_3 = a_4 = \dots$
- ☐ d. $0 = a_0 = a_2 = a_4 = a_6 = \dots$

At $x = 1$, the differential equation;

$$x^2(1 - x)^2 y'' + 2xy' + 4y = 0$$

has:

Select one:

- ☐ a. an irregular singular point
- ☐ b. a regular singular point
- ☐ c. none of the other answers
- ☐ d. an ordinary point

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3+9s}\right\} =$$

Select one:

- ☐ a. $\frac{1}{3}(\cos 3t - 1)$
- ☐ b. $-\frac{1}{9}(\cos 3t - 1)$
- ☒ c. $-\frac{1}{3}(\cos 3t - 1)$
- ☐ d. $\frac{1}{9}(\cos 3t - 1)$



Your answer is incorrect.

The correct answer is:

$$-\frac{1}{9}(\cos 3t - 1)$$

$$\int_1^7 \delta(t - 2) dt =$$

Select one:

☐ a. 0

☒ b. 1

☐ c. -2

☐ d. 2



$$\mathcal{L}\{t^4 e^{3t}\} =$$

Select one:

- ☐ a. $\frac{4}{(s-3)^5}$
- ☒ b. $\frac{24}{(s-3)^5}$
- ☐ c. $\frac{120}{(s-3)^6}$
- ☐ d. $\frac{5}{(s-3)^6}$

Clear my choice

$$\mathcal{L}\{u_{\pi}(t) \cos(t - 3\pi)\} =$$

Select one:

☐ a. $\frac{e^{-3\pi s}}{s^2+1}$

☐ b. $\frac{e^{-\pi s}}{s^2+1}$

☒ c. $\frac{se^{-\pi s}}{s^2+1}$

☐ d. $\frac{se^{-3\pi s}}{s^2+1}$

[Clear my choice](#)

$$\mathcal{L}^{-1}\left\{\frac{(s-2)e^{-2\pi s}}{s^2-4s+5}\right\} =$$

Select one:

- ☐ a. $-u_{2\pi}(t)e^{2t-4\pi}\cos t$
- ☐ b. $-u_{\pi}(t)e^{2t-2\pi}\cos t$
- ☒ c. $u_{2\pi}(t)e^{2t-4\pi}\cos t$
- ☐ d. $u_{\pi}(t)e^{2t-2\pi}\cos t$

[Clear my choice](#)

Suppose that $L\{e^{2t}f(t)\} = F_1(s)$, such that $L\{f(t)\} = \frac{s}{s+1}$, then $F_1(4) =$

- ☐ a. $\frac{3}{2}$
- ☐ b. $\frac{4}{3}$
- ☒ c. $\frac{3}{4}$
- ☐ d. $\frac{2}{3}$

[Clear my choice](#)

Consider $X' = \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} X$, such that the eigenvalues are $\lambda = \pm 2i$, then one of the two solutions is

- ☐ a. $\begin{pmatrix} 0 \\ 4 \end{pmatrix} \cos 2t - \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin 2t$
- ☐ b. $\begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t - \begin{pmatrix} 2 \\ 0 \end{pmatrix} \sin 2t$
- ☐ c. $\begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t - \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin 2t$
- ☐ d. $\begin{pmatrix} 0 \\ 4 \end{pmatrix} \cos 2t - \begin{pmatrix} 2 \\ 0 \end{pmatrix} \sin 2t$

If $L\{f(t)\} = \ln(s + 6) - \ln(s + 2)$, then
 $\int_0^\infty e^{-2t} f(t) dt =$

- ☐ a. $\ln 2$
- ☐ b. $\ln 12$
- ☐ c. $\ln 32$
- ☐ d. $-\ln 2$

If $A = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}$, then the eigenvalues of A are

- ☐ a. 3, 2
- ☐ b. 3, -3
- ☐ c. -3, -3
- ☒ d. 3, 3

[Clear my choice](#)

The $L\{t^4\} =$

☒ a. $\frac{24}{s^5}$

☐ b. $\frac{24}{s^4}$

☐ c. $\frac{12}{s^5}$

☐ d. $\frac{4}{s^5}$

[Clear my choice](#)